Alex Gower

Math & Algorithms Group, Nokia Bell Labs and Theoretical Condensed Matter Group, University of Cambridge

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Education

Nokia Bell Labs and University of Cambridge, Jesus College Sep 2022 – Present

PhD: Learning by Relaxation: Equilibrium Propagation for Neuromorphic Generative Modelling

Supervisors: Prof. Claudio Castelnovo (Cambridge), Dr Joachim Wabnig (Nokia Bell Labs)

University of Oxford, Lincoln College Oct 2018 – Jun 2022

MMathPhys: Master of Mathematical and Theoretical Physics, Distinction (73%)

BA: Physics, First Class (82%, Ranked 1st in Cohort of ∼200 in Second Year Examinations)

Publications

† *sole author*

† **Alex Gower**. PotentialEP: Backprop-Free Training of CIFAR-Scale Generative Models via Equilibrium Propagation. Submitted to *ICLR 2027*. arXiv:2610.14409

† **Alex Gower**. FlowEqProp: Training Flow Matching Generative Models with Gradient Equilibrium Propagation. In *Proc. of ICONS '26*, p60–66. IEEE Press, 2026. doi:10.1145/3822454.3822485 [**Best Paper Award**]

† **Alex Gower**. Learning at the Speed of Physics: Equilibrium Propagation on Oscillator Ising Machines. *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences (ML4PS)*. arXiv:2510.12934

† **Alex Gower**. How to Train an Oscillator Ising Machine using Equilibrium Propagation. In *Proc. of ICONS '25*, p229–234. IEEE Press, 2025. doi:10.1109/ICONS69015.2025.00042

Alex Gower, Oliver Hart, and Claudio Castelnovo. Saddles-to-Minima Topological Crossover and Glassiness in the Rubik's Cube. Submitted to *Nature Communications*. arXiv:2410.14552

M. Bristow, **A. Gower**, A. I. Coldea et al. Multiband Description of the Upper Critical Field of Bulk FeSe. *Physical Review B*, 108(18):184507, 2023.

Z. Zajicek, S. J. Singh, H. Jones, P. Reiss, M. Bristow, **A. Gower** et al. Drastic Effect of Impurity Scattering on the Electronic and Superconducting Properties of Cu-doped FeSe. *Physical Review B*, 105(11):115130, 2022.

Awards & Funding

- Best Paper Award, International Conference on Neuromorphic Systems (ICONS), 2026
- G-Research Grant (£2,000), 2025
- ICONS Student Travel Award ($1,000), 2025
- Best Poster Prize, Jesus College MCR Graduate Conference, 2024
- EPSRC CASE PhD Studentship in partnership with Nokia UK, 2022
- EPSRC Vacation Internship Award (£4,000), 2021
- Winton Capital Prize for 1st Place in Oxford Second Year Physics Examinations, 2020
- Lincoln College Lord Crewe Scholarship, Scholarship, Sidgwick Exhibition and Examination Prize, 2019–2022

Research Experience

Nokia Bell Labs and University of Cambridge Sep 2022 – Present
PhD Researcher, Math & Algorithms Group and Theoretical Condensed Matter Group

- Proved that a predictive coding network with a single added linear energy term relaxes to a state carrying the gradient of an Energy Matching potential, and trained it end to end by EP to 3.64 FID on unconditional CIFAR-10 (3.54 for backpropagation), the first backpropagation-free generative model at this quality (*PotentialEP*, submitted to ICLR 2027)
- Introduced Gradient Equilibrium Propagation (GradEP), extending EP to train energy gradients rather than energy minima, and demonstrated the first EP-trained flow matching generative model (FlowEqProp) — **Best Paper Award**, *ICONS '26*
- Developed a novel EP learning algorithm for Oscillator Ising Machines (MLP and CNN architectures), achieving SoTA neuromorphic accuracy on MNIST/FashionMNIST while maintaining hardware-constraint resilience (*ICONS 2025*, *NeurIPS 2025 ML4PS*); now being implemented on hardware with Nokia and TU Eindhoven
- Authored paper on rugged energy landscapes (submitted to *Nature Communications*) uncovering a saddles-to-minima topological crossover in the connectivity of stationary points that governs slow glassy dynamics, and its suppression (with fast dynamics recovered) upon the inclusion of additional 'swap moves' connections
- Engineered OIM performance for combinatorial optimisation (Max-SAT, Max-Cut, spin-glass models) using dimensional annealing and memory dynamics, and co-developed a GNN to learn optimal Max-SAT clause weights
- Built a multi-threaded Julia/PyTorch simulation pipeline with custom memory management, reducing runtime by 10x on distributed HPC clusters

Alan Turing Institute & UK Ministry of Justice Aug 2025 – Mar 2026
Machine Learning Technical Project Lead

- Built short-horizon risk forecasting models (LightGBM, XGBoost, EBM) using 1,500+ engineered features across 20M+ records, improving recall by 50% over prior baselines and achieving state-of-the-art short-term risk forecasting performance
- Built reduced-feature interpretable EBM models suitable for high-stakes decision-making, and surfaced actionable risk factors to inform frontline operations; built several LLM pipelines for semantic incident categorisation and case note surfacing

MARS Alignment Research Programme, Cambridge AI Safety Hub Dec 2024 – May 2025
Research Fellow

- Evaluated frontier LLM abilities in model compression, mechanistic interpretability and feature steering for ML R&D tasks using the InspectAI framework

Quantum Matter in High Magnetic Fields Group, University of Oxford Jun 2021 – Aug 2022
Undergraduate and Graduate Research Assistant (supervisor: Prof. Amalia Coldea)

- Co-authored two Physical Review B papers on multi-band superconductivity and impurity effects in FeSe, developing theoretical models and a MATLAB pipeline (tight-binding, multi-band, DFT) to extract band parameters from quantum oscillation and upper critical field experiments

ISIS Neutronics Group, Science and Technology Facilities Council Jun 2020 – Sep 2020
Nuclear Analysis Summer Student

- Benchmarked and extended the FETCH2 radiation transport codebase, correcting cross-section handling errors; presented results at an Imperial College conference

Selected Talks & Presentations

- International Conference on Neuromorphic Systems (ICONS) (opening contributed talk, Best Paper Award) — July 2026
- NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences (poster) — December 2025
- International Conference on Neuromorphic Systems (ICONS) 2025 (contributed talk, student travel award) — July 2025
- International Workshop on Ising Machines 2025 [IEEE NTC & TC-16] (poster) — May 2025
- American Physical Society March Meeting (contributed talk) — March 2025
- International Workshop on Ising Machines 2024 [IEEE NTC & TC-16] (oral presentation) — April 2024
- Current Research in Magnetism 2023: Magnetic Neuromorphic Computing [IOP] (poster) — September 2023

Teaching, Mentoring & Service

Theoretical Condensed Matter Group, University of Cambridge Oct 2024 – Present
Master's Project Supervisor

- Supervised two Master's dissertations investigating saddles-to-minima topological crossovers and high dimensional route suppression in kinetically constrained Ising models

Jesus College, University of Cambridge Oct 2022 – Oct 2025
Undergraduate Supervisor (Cambridge's small-group tutorial system; sole instructor)

- Prepared and taught fortnightly Mathematical Methods (second year) and Statistical Mechanics (third year, two successive years) supervisions to around 20 students in groups of 1–2

University of Cambridge Sep 2023
Organiser, Advanced Working Group on Slow Relaxation and Glassiness
NeurIPS Workshop on Machine Learning and the Physical Sciences (ML4PS) 2025–2026
Reviewer
Outreach 2018 – 2022
Lincoln College, LincZambia, STEPMaths

- Lincoln College ambassador and physics mentor for open days, interviews and new students; created a YouTube channel advising prospective Oxford physics applicants
- Taught computer skills and web development to 200+ students in Chipata, Zambia, and served as LincZambia charity president
- Produced 125 solution and animated visualisation videos and 4 mock papers for the PAT, ENGAA and STEP admissions exams

Additional Training & Skills

Training: MPhil in Machine Learning and Machine Intelligence (PhD auditor, Cambridge, 2023–present); ELLIS Cambridge Summer School on Probabilistic Machine Learning (selected participant, 2026); Cambridge AI Safety Hub Fellowship (2024–2025)

Programming: Python, Julia, MATLAB, C++, Java, R, FORTRAN

Libraries & Tools: PyTorch, JAX, NumPy, Hugging Face, Git, SLURM/HPC

Other Interests

Acrobatic Gymnastics (Cambridge University Double Blue), Jesus College Graduate Social Officer, Charity (LincZambia)